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EXPERIMENT - 5

System Calls in Linux

1. FORK System Call

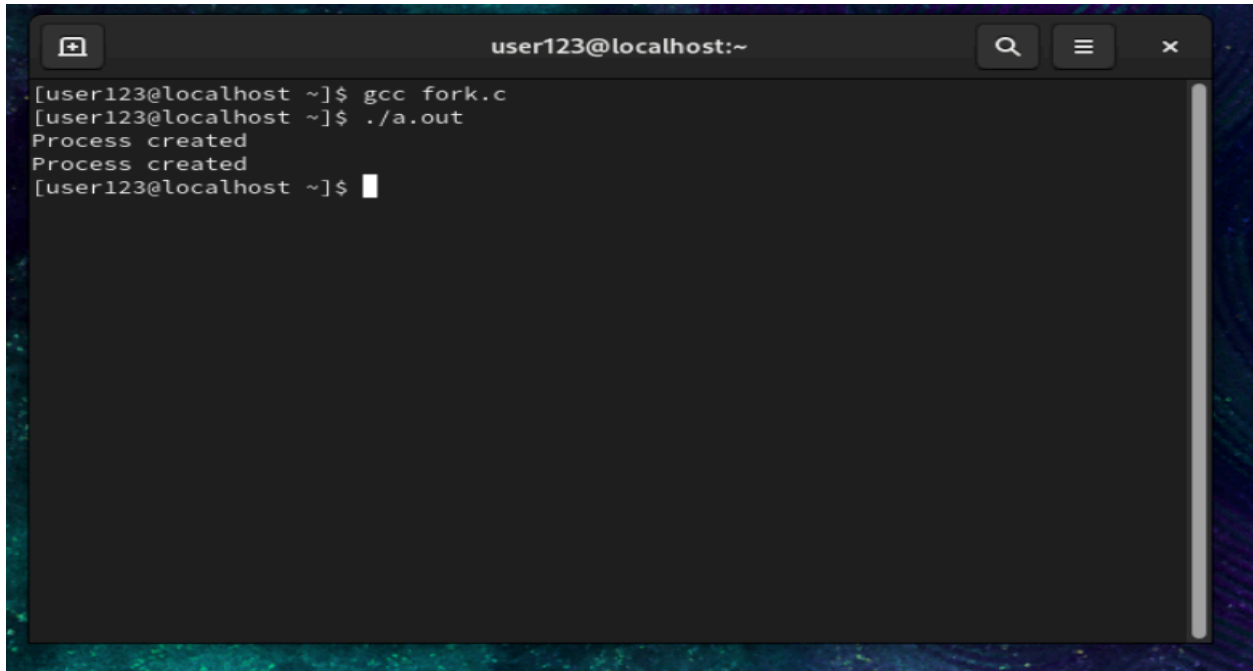
AIM:

To write and execute a C program using fork() system call.

CODE:

[illegible]

OUTPUT:



```
user123@localhost:~  
[user123@localhost ~]$ gcc fork.c  
[user123@localhost ~]$ ./a.out  
Process created  
Process created  
[user123@localhost ~]$
```

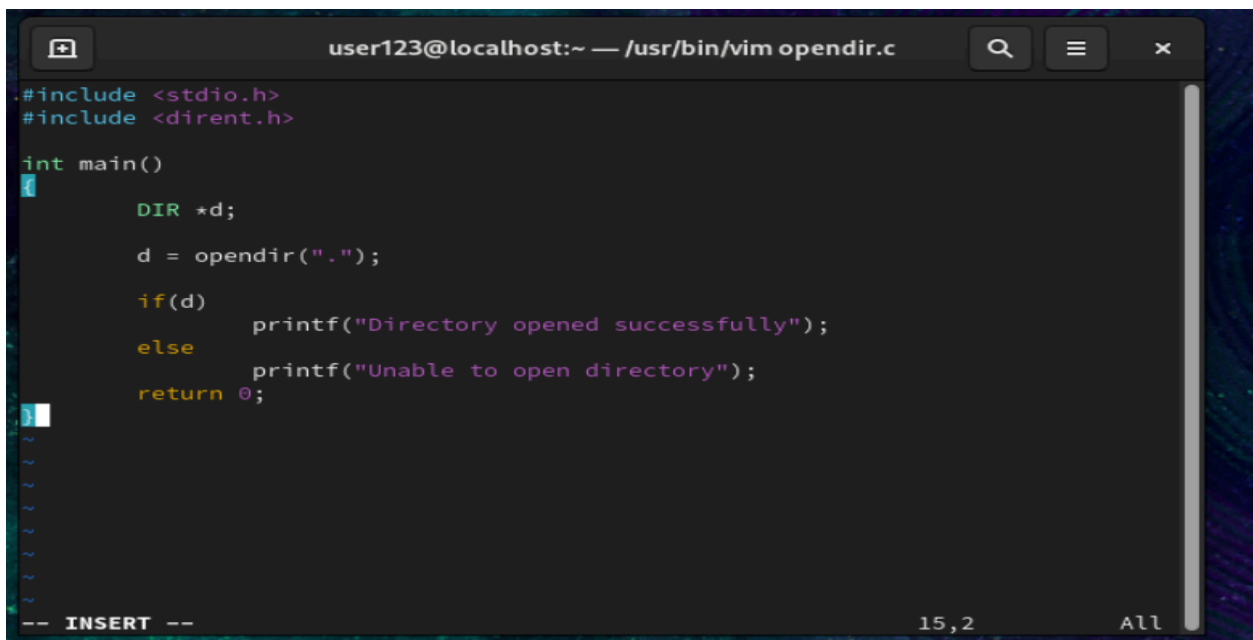
RESULT:

Thus, the fork() system call was executed successfully.

2. OPENDIR System Call

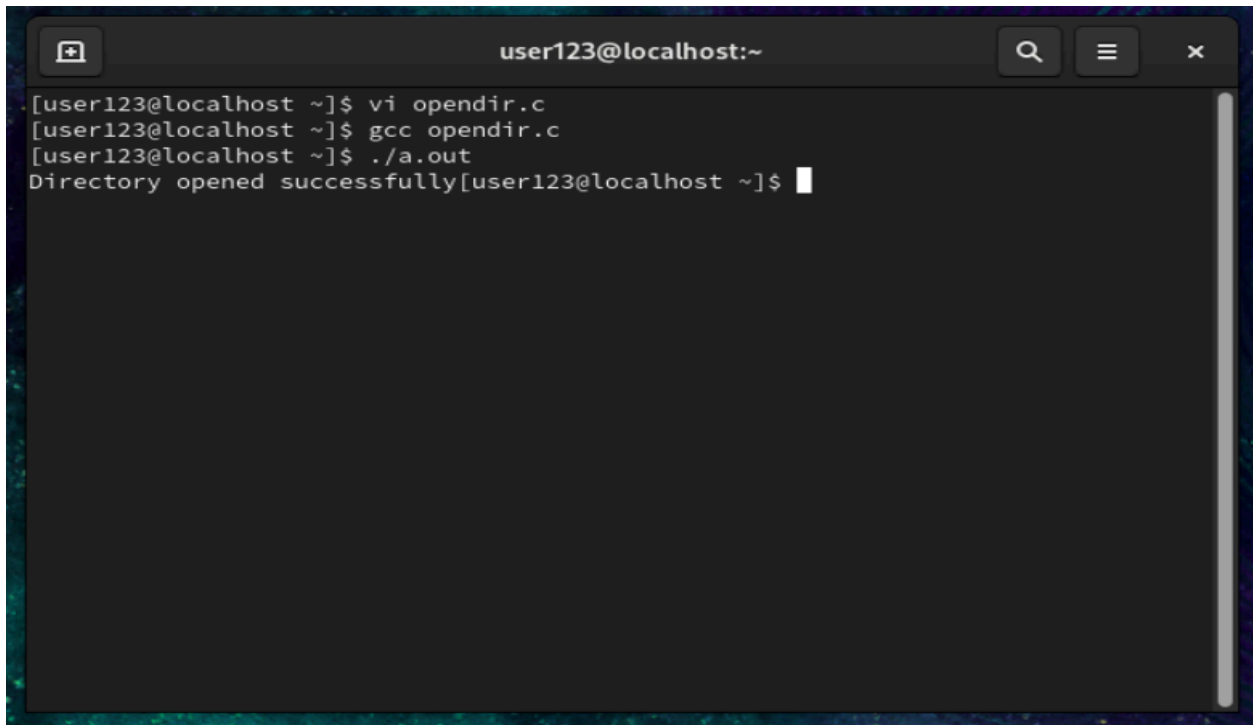
AIM: To write and execute a C program using opendir() system call.

CODE:



```
user123@localhost:~ — /usr/bin/vim opendir.c  
#include <stdio.h>  
#include <dirent.h>  
  
int main()  
{  
    DIR *d;  
    d = opendir(".");  
    if(d)  
        printf("Directory opened successfully");  
    else  
        printf("Unable to open directory");  
    return 0;  
}  
~  
~  
~  
~  
~  
~  
-- INSERT -- 15,2 All
```

OUTPUT:

A terminal window titled 'user123@localhost:~' with search, menu, and close buttons. It shows the following commands and output:

```
[user123@localhost ~]$ vi opendir.c
[user123@localhost ~]$ gcc opendir.c
[user123@localhost ~]$ ./a.out
Directory opened successfully[user123@localhost ~]$
```

RESULT:

Thus, the opendir() system call was executed successfully.

3. PID System Call

AIM: To write and execute a C program using getpid() system call.

CODE:

```

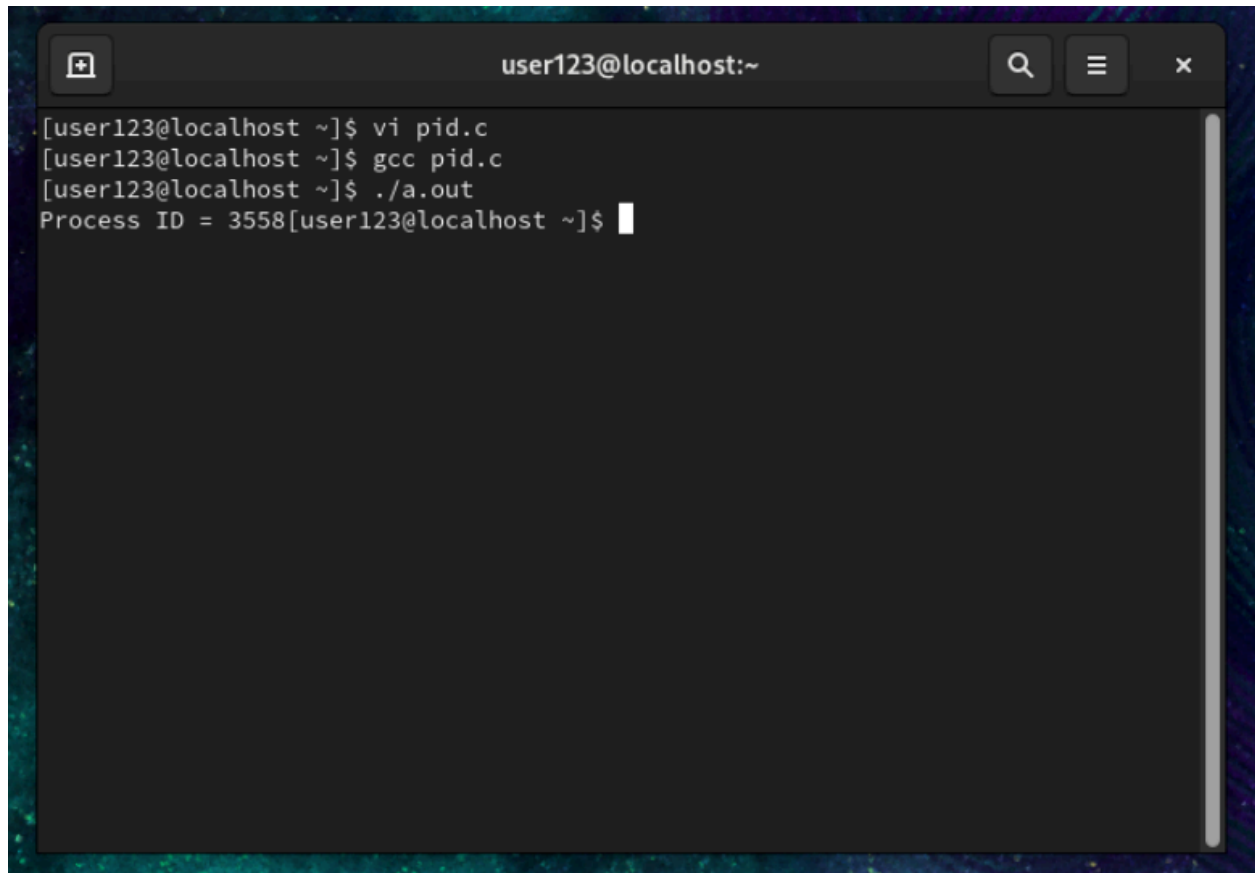
user123@localhost:~ — /usr/bin/vim pid.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process ID = %d", getpid());

    return 0;
}

-- INSERT --
9,2 All
```

OUTPUT:

A terminal window titled 'user123@localhost:~' with search, menu, and close buttons. It shows a sequence of commands: 'vi pid.c', 'gcc pid.c', and './a.out'. The output of the program is 'Process ID = 3558'.

```
[user123@localhost ~]$ vi pid.c
[user123@localhost ~]$ gcc pid.c
[user123@localhost ~]$ ./a.out
Process ID = 3558[user123@localhost ~]$
```

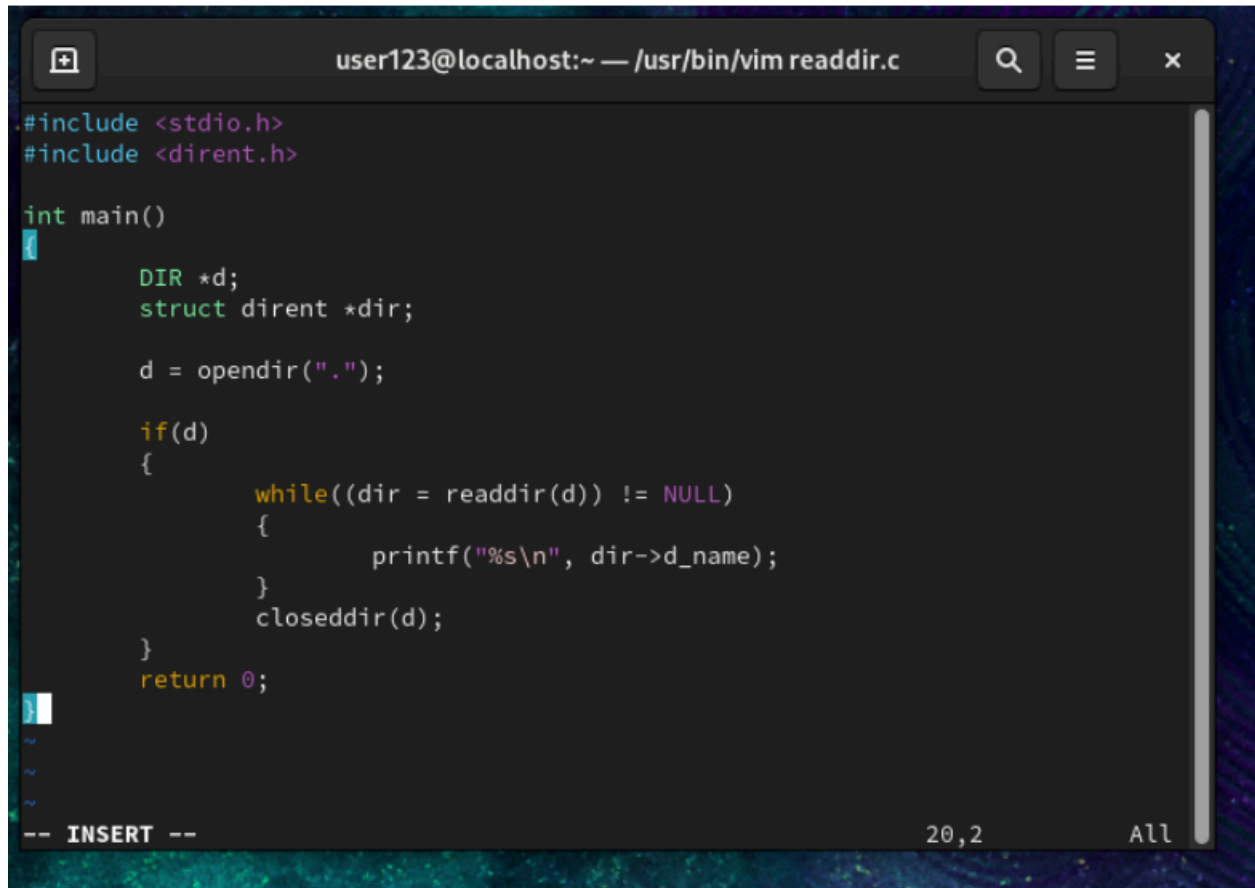
RESULT:

Thus, the getpid() system call was executed successfully.

4. READDIR System Call

AIM: To write and execute a C program using readdir() system call.

CODE:



```
user123@localhost:~ — /usr/bin/vim readdir.c

#include <stdio.h>
#include <dirent.h>

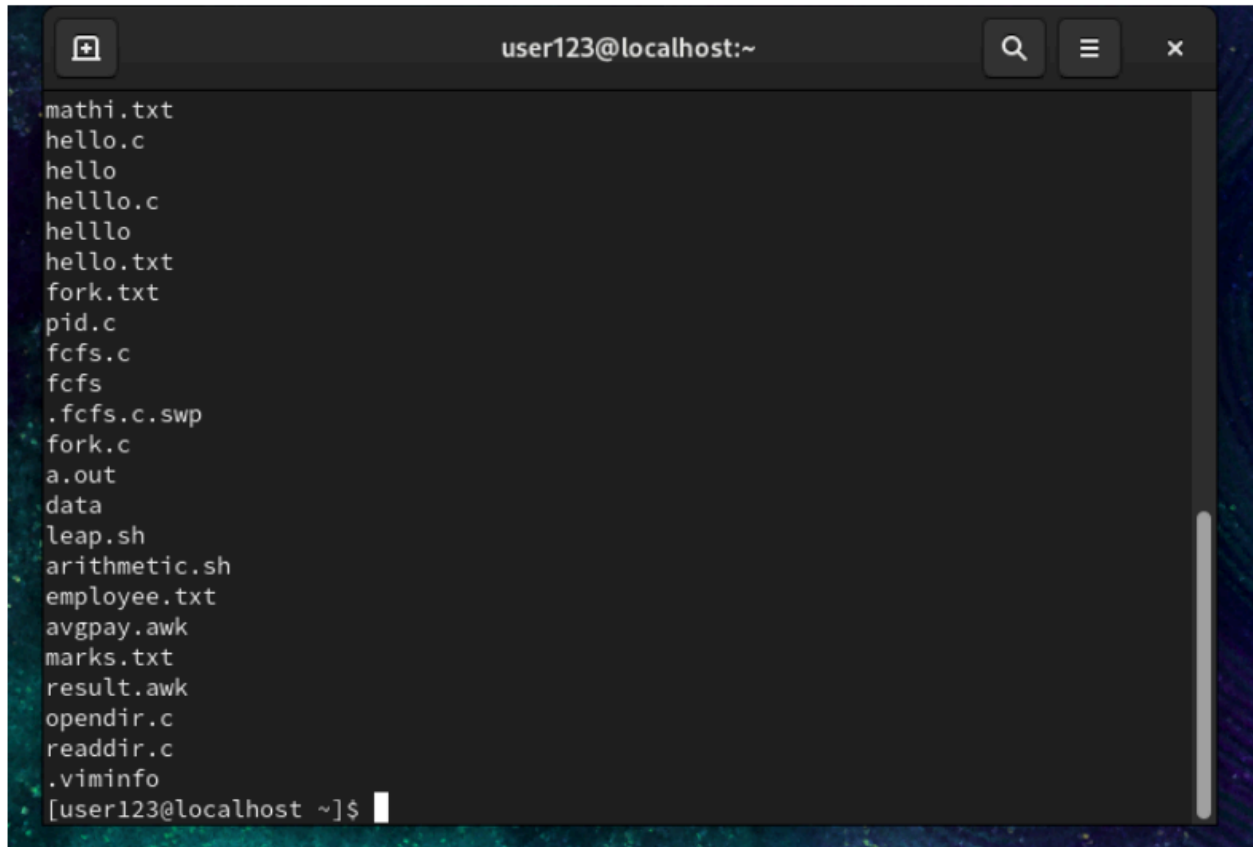
int main()
{
    DIR *d;
    struct dirent *dir;

    d = opendir(".");

    if(d)
    {
        while((dir = readdir(d)) != NULL)
        {
            printf("%s\n", dir->d_name);
        }
        closedir(d);
    }
    return 0;
}

~
~
~
-- INSERT --                               20,2      All
```

OUTPUT:

A terminal window titled 'user123@localhost:~' with search, menu, and close buttons. It displays a list of files in a directory: mathi.txt, hello.c, hello, hello.c, hello, hello.txt, fork.txt, pid.c, fcfs.c, fcfs, .fcfs.c.swp, fork.c, a.out, data, leap.sh, arithmetic.sh, employee.txt, avgpay.awk, marks.txt, result.awk, opendir.c, readdir.c, and .viminfo. The prompt '[user123@localhost ~]\$' is at the bottom.

```
user123@localhost:~  
mathi.txt  
hello.c  
hello  
hello.c  
hello  
hello.txt  
fork.txt  
pid.c  
fcfs.c  
fcfs  
.fcfs.c.swp  
fork.c  
a.out  
data  
leap.sh  
arithmetic.sh  
employee.txt  
avgpay.awk  
marks.txt  
result.awk  
opendir.c  
readdir.c  
.viminfo  
[user123@localhost ~]$
```

RESULT:

Thus, the readdir() system call was executed successfully.